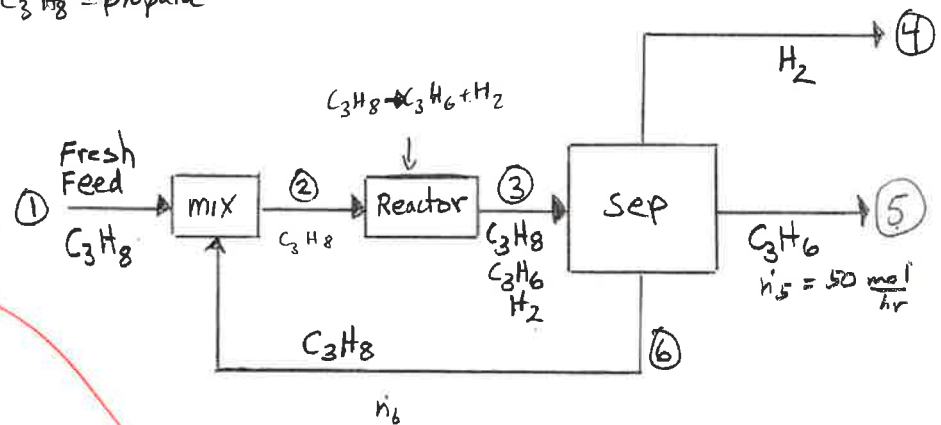


[1] (30 pts.) A process for propane dehydrogenation is shown in the figure. Assume that the only reaction taking place in the reactor is: $C_3H_8 \rightarrow C_3H_6 + H_2$ and there are no side reactions. There are six streams and the chemical species in each stream are labeled. The single-pass conversion for the reactor is 20% based on propane. The product flow rate is $\dot{n}_5 = 50 \text{ mol/hr}$ for Stream 5. Calculate all of the species molar flow rates and fill in your answers at the bottom of the page.

C_3H_8 = propane



DoF: $(4 + 4) - (5 + 5 + 5 + 5) = 2$

sep: $(4 + 0) - (1 + 2 + 1 + 3) = 2$

Mixer balance

$\dot{n}_1 + \dot{n}_6 = \dot{n}_2 \checkmark \rightarrow \dot{n}_1 + \dot{n}_2 - 50 = \dot{n}_2$
 $\dot{n}_1 = 50 \checkmark$

Reactor

$C_3H_8: \dot{n}_2 - \dot{F}_1 = \dot{n}_3, C_3H_8 \rightarrow \dot{n}_2 - 50 = \dot{n}_3, C_3H_8$

$C_3H_6: \dot{F}_1 = \dot{n}_3, C_3H_6 = 50 \text{ mol/hr} \checkmark$

$H_2: \dot{F}_1 = \dot{n}_3, H_2 = 50 \text{ mol/hr} \checkmark$

Separator

$C_3H_8: \dot{n}_3, C_3H_8 = \dot{n}_6 \rightarrow \dot{n}_2 - 50 = \dot{n}_6$

$C_3H_6: \dot{n}_3, C_3H_6 = \dot{n}_5 \rightarrow \dot{n}_3, C_3H_6 = 50 \text{ mol/hr}$

$H_2: \dot{n}_3, H_2 = \dot{n}_4$

use conversion

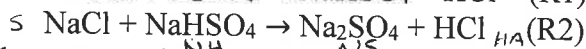
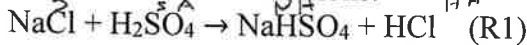
$\dot{n}_1 = 50 \checkmark \text{ mol/hr}, \dot{n}_2 = \text{ } \text{mol/hr}, \dot{n}_3, C_3H_8 = \text{ } \text{mol } C_3H_8/\text{hr},$

$\dot{n}_3, C_3H_6 = 50 \checkmark \text{ mol } C_3H_6/\text{hr}, \dot{n}_3, H_2 = 50 \checkmark \text{ mol } H_2/\text{hr}, \dot{n}_4 = \text{ } \text{mol/hr},$

$\dot{n}_6 = \text{ } \text{mol/hr}$

18

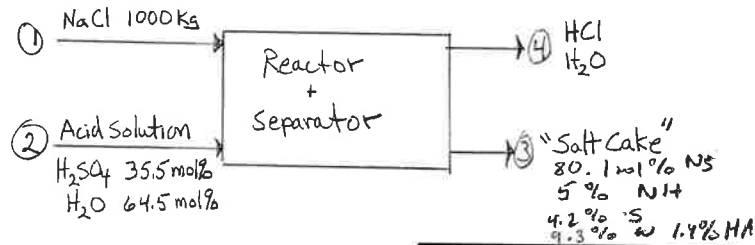
[2] (50 pts.) Salt (NaCl) and sulfuric acid (H_2SO_4) are reacted to produce hydrochloric acid (HCl) in a process represented by two chemical reactions:



For the process, there are two input streams: **Stream 1**:

1000 kg of pure NaCl and **Stream 2**: a 35.5 mol%

aqueous solution of sulfuric acid. There are two output streams: **Stream 3**: a "Salt Cake" consisting of 80.1 mol% of Na_2SO_4 , 5.0 mol% of $NaHSO_4$, 4.2 mol% of NaCl, 9.3 mol% of H_2O , and 1.4 mol% of HCl; and **Stream 4**: a vapor stream of HCl and H_2O .



Species	Molar Mass
Na_2SO_4	142.1
$NaHSO_4$	120.1
NaCl	58.5
H_2O	18.0
HCl	36.5
H_2SO_4	98.1

(a) Calculate the mole of NaCl entering the process in kmol. (5 pts.)

$$\frac{kg}{kmol} = \frac{g}{mol} \quad 1000 \text{ kg NaCl} \times \frac{1 \text{ mol}}{58.5 \text{ g}} \times \frac{1 \text{ kmol}}{1 \text{ kg}} = 17.09 \text{ kmol}$$

$$1000 \text{ kg} \times \frac{1000 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol}}{58.5 \text{ g}} =$$

$$n_1 = 17.09 \text{ kmol}$$

(b) Write species mole balance equations and/or atomic balance equations. The problem can be solved by either method. (20 pts.)

① S: $n_1 = \dot{F}_1 - \dot{F}_2 = \dot{n}_3 \times 3.258$
 $17.09 - \dot{F}_1 - \dot{F}_2 = \dot{n}_3 (0.042)$
 $\dot{F}_1 - \dot{F}_2 = 2.457 \dot{n}_3 - 17.09$
 $-3.258 \dot{n}_3 - (-3.258 \dot{n}_3 - 6.0 \dot{n}_3) = 2.457 \dot{n}_3 - 17.09$
 $3.258 \dot{n}_3$
 $34.83 \dot{n}_2 = 3.258 (124.47)$
 $\dot{n}_2 = 11.64$

② SA: $-\dot{F}_1 + \dot{n}_2 \times 0.355 = 0 \rightarrow \dot{n}_2 (0.355) = \dot{F}_1 \rightarrow 34.83 \dot{n}_2 = \dot{F}_1$
W: $\dot{n}_2 \times 0.645 = \dot{n}_4 \times 0.18 + \dot{n}_3 (0.093)$
 $11.61 \dot{n}_2 = \dot{n}_4 \times 0.18 + 1.674 \dot{n}_3$

③ NS: $-\dot{F}_2 = \dot{n}_3 \times 0.801 \rightarrow -\dot{F}_2 = \dot{n}_3 (0.801) \rightarrow -\dot{F}_2 = 113.75 \dot{n}_3$

NH: $\dot{F}_1 - \dot{F}_2 = \dot{n}_3 \times 0.05 \rightarrow \dot{F}_1 - \dot{F}_2 = \dot{n}_3 (0.05) \rightarrow \dot{F}_2 = \dot{F}_1 - 6.0 \dot{n}_3$

HA: $\dot{F}_1 + \dot{F}_2 = \dot{n}_3 \times 0.014 \rightarrow \dot{F}_1 + \dot{F}_2 = \dot{n}_3 (0.014) \rightarrow \dot{F}_1 + \dot{F}_1 - 6.0 \dot{n}_3 = 0.511 \dot{n}_3$

$\dot{n}_3 = \dot{n}_3 \times 3.258 + \dot{n}_3 \times 0.042 + \dot{n}_3 \times 0.05 + \dot{n}_3 \times 0.093 + \dot{n}_3 \times 0.801$

$$2 \dot{F}_1 = 6.5 \dot{n}_3$$

$$\dot{F}_1 = 3.258 \dot{n}_3$$

$$\dot{n}_2 = \dot{n}_2 \times 0.355 + \dot{n}_2 \times 0.645$$

20

PROBLEM [2] CONTINUED

(c) Calculate the molar amounts of Streams 2, 3, and 4; and the mole fraction of HCl in Stream 4. (15 pts.)

$$\dot{n}_3 = 0.042(58.5) + 0.801(142.1) + 0.05(120.1) + 0.093(18) + 0.014(36.5) = 124.47 \text{ kmol}$$

$$\dot{n}_2 = 0.355(98.1) + 0.645(18) = 46.44 \text{ kmol}$$

$$n_1 + n_2 = n_3 + n_4$$

$$n_4 = n_1 + n_2 - n_3$$

$$n_4 = 17.09 + 46.44 - 124.47 =$$

5

$$n_2 = 46.44 \text{ kmol}, n_3 = 124.47 \text{ kmol}, n_4 = \text{_____ kmol}, x_{4,\text{HCl}} = \text{_____}$$

(d) Check the mass balance for the process. That is, show that mass in equals mass out. (10 pts.)

$$m_{in} = m_{out}$$

2

$$m_1 + m_{2,\text{H}_2\text{SO}_4} + m_{2,\text{H}_2\text{O}} = m_{4,\text{HCl}} + m_{4,\text{H}_2\text{O}} + m_3$$

$$\begin{array}{r|l} 1 & 18 \\ \hline 2 & 32 \\ \hline 3 & 10 \\ \hline \hline & 60 \end{array}$$

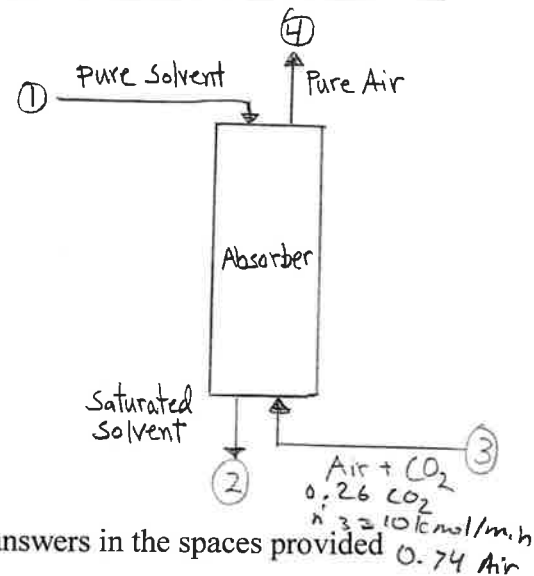
(32)

NAME Su Nuam

[3] (20 pts.) Carbon dioxide (CO₂) can be removed from air using an absorption process. In the figure, **Stream 1** is a pure solvent stream and **Stream 3** is an air/CO₂ mixture containing 26 mol% CO₂ at a flow rate of 10 kmol/min. The operating temperature and pressure is 50°C and 1 atm. Assume Stream 2 and Stream 3 are in equilibrium at 50°C.

Equilibrium Data at 50°C:

Partial Pressure of CO ₂ (mmHg) in Air/CO ₂	Liquid Concentration (moles of CO ₂ /mole of solvent)
668.4	0.680
242.3	0.562
183.8	0.548
71.0	0.489
10.2	0.302



Calculate the molar flow rates and compositions of all streams. Write your answers in the spaces provided below.

$$\dot{n}_1 + \dot{n}_3 = \dot{n}_2 + \dot{n}_4 \quad \checkmark$$

$$PV = nRT$$

$$\dot{n}_{1, \text{solvent}} = \dot{n}_{2, \text{sat. sol.}} \quad \checkmark$$

$$\text{Air: } \dot{n}_{3, \text{Air}} = \dot{n}_4$$

$$0.74 (10 \frac{\text{kmol}}{\text{min}}) = \dot{n}_4$$

$$7.4 = \dot{n}_4 \quad \checkmark$$

$$\text{CO}_2: \dot{n}_{3, \text{CO}_2} = 0.26 (10 \frac{\text{kmol}}{\text{min}}) = 2.6 \frac{\text{kmol}}{\text{min}} \quad \checkmark$$

$$y_i = \frac{p_i^{\text{sat}}}{P} x_i \rightarrow x_{\text{CO}_2}$$

use data

$$\dot{n}_1 = \underline{\hspace{2cm}} \text{ molS/min, } \dot{n}_2 = \underline{\hspace{2cm}} \text{ mol/min, } \dot{n}_4 = \underline{7.4} \text{ molAir/min,}$$

$$x_{\text{CO}_2} = \underline{\hspace{2cm}} \text{ molCO}_2/\text{mol}$$

10